



Real Python Pocket Reference

Visit realpython.com to turbocharge your Python learning with in-depth tutorials, real-world examples, and expert guidance.

Getting Started

Follow these guides to kickstart your Python journey:

- realpython.com/what-can-i-do-with-python
- realpython.com/installing-python
- realpython.com/python-first-steps

Start the Interactive Shell

```
$ python
```

Quit the Interactive Shell

```
>>> exit()
```

Run a Script

```
$ python my_script.py
```

Run a Script in Interactive Mode

```
$ python -i my_script.py
```

Learn More on realpython.com/search:

[interpreter](#) · [run a script](#) · [command line](#)

Comments

- Always add a space after the `#`
- Use comments to explain “why” of your code

Write Comments

```
# This is a comment
# print("This code will not run.")
print("This will run.") # Comments are ignored by Python
```

Learn More on realpython.com/search:

[comment](#) · [documentation](#)

Data Types

- Python is dynamically typed
- Use `None` to represent missing or optional values
- Use `type()` to check object type
- Check for a specific type with `isinstance()`
- `issubclass()` checks if a class is a subclass

Type Investigation

```
type(42) # <class 'int'>
type(3.14) # <class 'float'>
type("Hello") # <class 'str'>
type(True) # <class 'bool'>
type(None) # <class 'NoneType'>

isinstance(3.14, float) # True
issubclass(int, object) # True - everything inherits from object
```

Type Conversion

```
int("42") # 42
float("3.14") # 3.14
str(42) # "42"
bool(1) # True
list("abc") # ["a", "b", "c"]
```

Learn More on realpython.com/search:

[data types](#) · [type checking](#) · [isinstance](#) · [issubclass](#)

Variables & Assignment

- Variables are created when first assigned
- Use descriptive variable names
- Follow `snake_case` convention

Basic Assignment

```
name = "Leo" # String
age = 7 # Integer
height = 5.6 # Float
is_cat = True # Boolean
flaws = None # None type
```

Parallel & Chained Assignments

```
x, y = 10, 20 # Assign multiple values
a = b = c = 0 # Give same value to multiple variables
```

Augmented Assignments

```
counter += 1
numbers += [4, 5]
permissions |= write
```

Learn More on realpython.com/search:

[variables](#) · [assignment operator](#) · [walrus operator](#)

Strings

- It's recommended to use double-quotes for strings
- Use `"\n"` to create a line break in a string
- To write a backslash in a normal string, write `"\\"`

Creating Strings

```
single = 'Hello'
double = "World"
multi = """Multiple
line string"""
```

String Operations

```
greeting = "me" + "ow!" # "meow!"
repeat = "Meow!" * 3 # "Meow!Meow!Meow!"
length = len("Python") # 6
```

String Methods

```
"a".upper() # "A"
"A".lower() # "a"
"a ".strip() # "a"
"abc".replace("bc", "ha") # "aha"
"a b".split() # ["a", "b"]
"-.".join(["a", "b"]) # "a-b"
```

String Indexing & Slicing

```
text = "Python"
text[0] # "P" (first)
text[-1] # "n" (last)
text[1:4] # "yth" (slice)
text[:3] # "Pyt" (from start)
text[3:] # "hon" (to end)
text[::-2] # "Pto" (every 2nd)
text[::-1] # "nohtyP" (reverse)
```

String Formatting

```
# f-strings
name = "Aubrey"
age = 2
f"Hello, {name}!" # "Hello, Aubrey!"
f"{name} is {age} years old" # "Aubrey is 2 years old"
f"Debug: {age}" # "Debug: age=2"

# Format method
template = "Hello, {name}! You're {age}."
template.format(name="Aubrey", age=2) # "Hello, Aubrey! You're 2."
```

Raw Strings

```
# Normal string with an escaped tab
"This is:\tCool." # "This is: Cool."

# Raw string with escape sequences
r"This is:\tCool." # "This is:\tCool."
```

Learn More on realpython.com/search:

[strings](#) · [string methods](#) · [slice notation](#) · [raw strings](#)

Numbers & Math

Arithmetic Operators

```
10 + 3 # 13
10 - 3 # 7
10 * 3 # 30
10 / 3 # 3.3333333333333335
10 // 3 # 3
10 % 3 # 1
2 ** 3 # 8
```

Useful Functions

```
abs(-5) # 5
round(3.7) # 4
round(3.14159, 2) # 3.14
min(3, 1, 2) # 1
max(3, 1, 2) # 3
sum([1, 2, 3]) # 6
```

Learn More on realpython.com/search:

[math](#) · [operators](#) · [built in functions](#)

Conditionals

- Python uses indentation for code blocks
- Use 4 spaces per indentation level

If-Elif-Else

```
if age < 13:
    category = "child"
elif age < 20:
    category = "teenager"
else:
    category = "adult"
```

Comparison Operators

```
x == y # Equal to
x != y # Not equal to
x < y # Less than
x <= y # Less than or equal
x > y # Greater than
x >= y # Greater than or equal
```

Logical Operators

```
if age >= 18 and has_car:
    print("Roadtrip!")

if is_weekend or is_holiday:
    print("No work today.")

if not is_raining:
    print("You can go outside.")
```

Learn More on realpython.com/search:

[conditional statements](#) · [operators](#) · [truthy falsy](#)

Loops

- `range(5)` generates 0 through 4
- Use `enumerate()` to get index and value
- `break` exits the loop, `continue` skips to next
- Be careful with `while` to not create an infinite loop

For Loops

```
# Loop through range
for i in range(5):    # 0, 1, 2, 3, 4
    print(i)

# Loop through collection
fruits = ["apple", "banana"]
for fruit in fruits:
    print(fruit)

# With enumerate for index
for i, fruit in enumerate(fruits):
    print(f"{i}: {fruit}")
```

While Loops

```
while True:
    user_input = input("Enter 'quit' to exit: ")
    if user_input == "quit":
        break
    print(f"You entered: {user_input}")
```

Loop Control

```
for i in range(10):
    if i == 3:
        continue # Skip this iteration
    if i == 7:
        break # Exit loop
    print(i)
```

Learn More on realpython.com/search:

[for loop](#) · [while loop](#) · [enumerate](#) · [control flow](#)

Functions

- Define functions with `def`
- Always use `()` to call a function
- Add `return` to send values back
- Create anonymous functions with the `lambda` keyword

Defining Functions

```
def greet():
    return "Hello!"

def greet_person(name):
    return f"Hello, {name}!"

def add(x, y=10):    # Default parameter
    return x + y
```

Calling Functions

```
greet()           # "Hello!"
greet_person("Bartosz") # "Hello, Bartosz"
add(5, 3)         # 8
add(7)           # 17
```

Return Values

```
def get_min_max(numbers):
    return min(numbers), max(numbers)

minimum, maximum = get_min_max([1, 5, 3])
```

Useful Built-in Functions

```
callable() # Checks if an object can be called as a function
dir()      # Lists attributes and methods
globals()  # Get a dictionary of the current global symbol table
hash()     # Get the hash value
id()       # Get the unique identifier
locals()   # Get a dictionary of the current local symbol table
repr()     # Get a string representation for debugging
```

Lambda Functions

```
square = lambda x: x**2
result = square(5) # 25

# With map and filter
numbers = [1, 2, 3, 4]
squared = list(map(lambda x: x**2, numbers))
evens = list(filter(lambda x: x % 2 == 0, numbers))
```

Learn More on realpython.com/search:

[define functions](#) · [return multiple values](#) · [lambda](#)

Classes

- Classes are blueprints for objects
- You can create multiple instances of one class
- You commonly use classes to encapsulate data
- Inside a class, you provide methods for interacting with the data
- `__init__()` is the constructor method
- `self` refers to the instance

Defining Classes

```
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def bark(self):
        return f"{self.name} says Woof!"

# Create instance
my_dog = Dog("Frieda", 3)
print(my_dog.bark()) # Frieda says Woof!
```

Class Attributes & Methods

```
class Cat:
    species = "Felis catus" # Class attribute

    def __init__(self, name):
        self.name = name # Instance attribute

    def meow(self):
        return f"{self.name} says Meow!"

    @classmethod
    def create_kitten(cls, name):
        return cls(f"Baby {name}")
```

Inheritance

```
class Animal:
    def __init__(self, name):
        self.name = name

    def speak(self):
        pass

class Dog(Animal):
    def speak(self):
        return f"{self.name} barks!"
```

Learn More on realpython.com/search:

[object oriented programming](#) · [classes](#)

Exceptions

- When Python runs and encounters an error, it creates an exception
- Use specific exception types when possible
- `else` runs if no exception occurred
- `finally` always runs, even after errors

Try-Except

```
try:
    number = int(input("Enter a number: "))
    result = 10 / number
except ValueError:
    print("That's not a valid number!")
except ZeroDivisionError:
    print("Cannot divide by zero!")
else:
    print(f"Result: {result}")
finally:
    print("Calculation attempted")
```

Common Exceptions

```
ValueError # Invalid value
TypeError # Wrong type
IndexError # List index out of range
KeyError # Dict key not found
FileNotFoundError # File doesn't exist
```

Raising Exceptions

```
def validate_age(age):
    if age < 0:
        raise ValueError("Age cannot be negative")
    return age
```

Learn More on realpython.com/search:

[exceptions](#) · [errors](#) · [debugging](#)

Collections

- A collection is any container data structure that stores multiple items
- If an object is a collection, then you can loop through it
- Strings are collections, too
- Use `len()` to get the size of a collection
- You can check if an item is in a collection with the `in` keyword
- Some collections may look similar, but each data structure solves specific needs

Lists

```
# Creating lists
empty = []
nums = [5]
mixed = [1, "two", 3.0, True]

# List methods
nums.append("x") # Add to end
nums.insert(0, "y") # Insert at index 0
nums.extend(["z", 5]) # Extend with iterable
nums.remove("x") # Remove first "x"
last = nums.pop() # Pop returns last element

# List indexing and checks
fruits = ["banana", "apple", "orange"]
fruits[0] # "banana"
fruits[-1] # "orange"
"apple" in fruits # True
len(fruits) # 3
```

Tuples

```
# Creating tuples
point = (3, 4)
single = (1,) # Note the comma!
empty = ()

# Basic tuple unpacking
point = (3, 4)
x, y = point
x # 3
y # 4

# Extended unpacking
first, *rest = (1, 2, 3, 4)
first # 1
rest # [2, 3, 4]
```

Sets

```
# Creating Sets
a = {1, 2, 3}
b = set([3, 4, 4, 5])

# Set Operations
a | b      # {1, 2, 3, 4, 5}
a & b      # {3}
a - b      # {1, 2}
a ^ b      # {1, 2, 4, 5}
```

Dictionaries

```
# Creating Dictionaries
empty = {}
pet = {"name": "Leo", "age": 42}

# Dictionary Operations
pet["sound"] = "Purr!" # Add key and value
pet["age"] = 7         # Update value
age = pet.get("age", 0) # Get with default
del pet["sound"]        # Delete key
pet.pop("age")          # Remove and return

# Dictionary Methods
pet = {"name": "Frieda", "sound": "Bark!"}
pet.keys()      # dict_keys(['name', 'sound'])
pet.values()     # dict_values(['Frieda', 'Bark!'])
pet.items()      # dict_items([('name', 'Frieda'), ('sound', 'Bark!')])
```

Learn More on realpython.com/search:

list · tuple · set · dictionary · indexing · unpacking

Comprehensions

- You can think of comprehensions as condensed **for** loops
- Comprehensions are faster than equivalent loops

List Comprehensions

```
# Basic
squares = [x**2 for x in range(10)]

# With condition
evens = [x for x in range(20) if x % 2 == 0]

# Nested
matrix = [[i*j for j in range(3)] for i in range(3)]
```

Other Comprehensions

```
# Dictionary comprehension
word_lengths = {word: len(word) for word in ["hello", "world"]}

# Set comprehension
unique_lengths = {len(word) for word in ["who", "what", "why"]}

# Generator expression
sum_squares = sum(x**2 for x in range(1000))
```

Learn More on realpython.com/search:

comprehensions · data structures · generators

File I/O

File Operations

```
# Read an entire file
with open("file.txt", mode="r", encoding="utf-8") as file:
    content = file.read()

# Read a file line by line
with open("file.txt", mode="r", encoding="utf-8") as file:
    for line in file:
        print(line.strip())

# Write a file
with open("output.txt", mode="w", encoding="utf-8") as file:
    file.write("Hello, World!\n")

# Append to a File
with open("log.txt", mode="a", encoding="utf-8") as file:
    file.write("New log entry\n")
```

Learn More on realpython.com/search:

files · context manager · pathlib

Imports & Modules

- Prefer explicit imports over **import ***
- Use aliases for long module names
- Group imports: standard library, third-party libraries, user-defined modules

Import Styles

```
# Import entire module
import math
result = math.sqrt(16)

# Import specific function
from math import sqrt
result = sqrt(16)

# Import with alias
import numpy as np
array = np.array([1, 2, 3])

# Import all (not recommended)
from math import *
```

Package Imports

```
# Import from package
import package.module
from package import module
from package.subpackage import module

# Import specific items
from package.module import function, Class
from package.module import name as alias
```

Learn More on realpython.com/search:

import · modules · packages

Virtual Environments

- Virtual Environments are often called “venv”
- Use venvs to isolate project packages from the system-wide Python packages

Create Virtual Environment

```
$ python -m venv .venv
```

Activate Virtual Environment (Windows)

```
PS> .venv\Scripts\activate
```

Activate Virtual Environment (Linux & macOS)

```
$ source .venv/bin/activate
```

Deactivate Virtual Environment

```
(.venv) $ deactivate
```

Learn More on realpython.com/search:

virtual environment · venv

Packages

- The official third-party package repository is the Python Package Index (PyPI)

Install Packages

```
$ python -m pip install requests
```

Save Requirements & Install from File

```
$ python -m pip freeze > requirements.txt
$ python -m pip install -r requirements.txt
```

Related Tutorials

- Installing Python Packages
- Requirements Files in Python Projects

Miscellaneous

Truthy	Falsy
-42	0
3.14	0.0
"John"	""
[1, 2, 3]	[]
("apple", "banana")	()
{"key": None}	{}
	None

Pythonic Constructs

```
# Swap variables
a, b = b, a

# Flatten a list of lists
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
flat = [item for sublist in matrix for item in sublist]

# Remove duplicates
unique_unordered = list(set(my_list))

# Remove duplicates, preserve order
unique = list(dict.fromkeys(my_list))

# Count occurrences
from collections import Counter

counts = Counter(my_list)
```

Learn More on realpython.com/search:

counter · tricks

Do you want to go deeper on any topic in the Python curriculum?

At Real Python you can immerse yourself in any topic. Level up your skills effectively with curated resources like:

- Learning paths
- Video courses
- Written tutorials
- Interactive quizzes
- Podcast interviews
- Reference articles

Continue your learning journey and become a Python expert at realpython.com/start-here 🧠🔗